DONI VISHWA VARDHAN

AI/ML Engineer

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ACADEMIC DETAILS

Indian Institute of Technology Delhi *2026*

B.Tech in Biochemical Engineering & Biotechnology — CGPA: 7

EXPERIENCE

Management Trainee – Program Manager | Basic Enterprises *June 2026 – Present*

Products: CleverTap • Agent Flash • Invoice Rejection Tracker • Agent Retention Strategies

- Designed, built, and deployed a Retrieval-Augmented Generation (RAG) AI agent for Agent Support and Call Center operations, taking it from prototype to internal production use.
- Led end-to-end product development, from Figma wireframes and requirement scoping to feature delivery, using data-driven insights to solve user problems.
- Analyzed CleverTap product analytics to identify user drop-offs and pain points, prioritized the roadmap, and collaborated with engineering and operations teams to define success metrics and deliver data-informed solutions.

PROJECTS

MCP Gateway Platform | GitHub

TypeScript, Express, PostgreSQL, React, Kubernetes (k3s/k3d), Helm, Docker, GitHub Actions

- Built a team **MCP gateway** that puts 3 upstream servers (WooCommerce, FX, notes) behind one /mcp endpoint, with <server>__<tool> namespacing to prevent name collisions, 10-second health checks, per-user revocable API keys and an audit log of every tool call.
- Secured it with encrypted upstream credentials, **SSRF-guarded** server registration, admin/member roles, and Kubernetes Secrets, NetworkPolicies and probes. Shipped Helm charts for **local k3d and VPS k3s**, with multi-arch CI images deployed by commit SHA and one-command rollback.
- Added a React dashboard and an LLM chat agent (**gpt-oss-20b on Groq**) that runs a tool loop as the logged-in user. Backed by **363 automated tests in CI**. Replaces per-person setup of each server with one URL and one key, and gives admins visibility and instant access revocation.

Support Triage MCP Server — M8ven Verification |Deployment

Model Context Protocol (MCP), FAISS, Python,FastAPI

- Built an **MCP server** and a **FastAPI** web dashboard over the same triage tools (classify, draft reply, duplicate check, file and update cases, queue Q&A), where filing a case runs triage live. Added **rate limiting**, **input validation** and **HTML escaping** for a public demo.
- Grounded **gpt-oss-20b (Groq)** classification in similar resolved tickets. On 200 held-out Banking77 queries grouped into 9 **routing** queues, **accuracy** rose from **79.0% zero-shot to 93.5%** (95% CI 89.2 to 96.2), against 68.5% for retrieval alone. Built **duplicate detection** whose F1 rose from **0.71 (hashing) to 0.82 (BGE-small embeddings)** on 24 labeled pairs.
- Wrote 120+ pytest tests with GitHub Actions CI and an evaluation harness that reports confidence intervals.

Financial Intelligence Agent | GitHub

Python, LangGraph, Qdrant, PostgreSQL, FastAPI, Docker, Ollama

- Built a RAG-based financial research agent(Parsing, Data Cleaning, Chunking, Embedding) to analyze annual reports and financial documents using open-source LLMs (via **Ollama**), **LangGraph**, and **Qdrant** vector store with hybrid **dense+BM25** retrieval.
- Developed tool-using agent workflows (PostgreSQL **text-to-SQL**, financial calculations, document retrieval) exposed as APIs via FastAPI, with plan-then-execute orchestration and **sqlglot**-validated SQL execution.
- Implemented an **LLM-as-judge** evaluation harness scoring **answer correctness = 4.4/5**, **groundedness = 4.6/5**, and **routing accuracy = 94.6%**, with a defense-in-depth SQL safety layer (validated + read-only) and Dockerized deployment.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, Dart

ML / Deep Learning: Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost, Hugging Face Transformers, NLP, Numpy, Pandas

GenAI & LLM Engineering: LangChain, LangGraph, RAG pipelines, Prompt Engineering, Vector DBs (Chroma, FAISS, Qdrant), Ollama, Fine-tuning (LoRA/QLoRA, PEFT, trl, bitsandbytes), Guardrails (Llama Guard), Gateways (LiteLLM), Model Context Protocol (MCP)

MLOps & Deployment: FastAPI, Docker, Kubernetes, MLflow, Azure ML, AWS SageMaker, Vertex AI, CI/CD (GitHub Actions), Data Bricks, React, Vercel, Next JS, REST APIs

Cloud: Microsoft Azure, AWS,GCP

Tools: Git, VS Code, MS Excel, MS Office